

CHE 313 Spring 2025 Self-Study Report

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Course Comments / Plans

The course requires students to develop a rigorous understanding of mass and energy balances to formulate governing equations for separation and transport processes. Assessment of student performance indicates that a significant portion of the class continues to experience difficulty in systematically applying conservation principles, particularly when translating physical separation processes into mathematical representations.

Common challenges include:

- Incomplete or incorrect identification of system boundaries
- Errors in accounting for inlet/outlet streams and accumulation terms

To address these gaps, instructional emphasis has been placed on structured problem formulation. Students are guided to:

1. Construct clear process flow diagrams or sketches
2. Explicitly define system boundaries and assumptions
3. Identify all relevant input/output streams before writing governing equations

Worked examples and in-class problem-solving sessions have been incorporated to reinforce these steps across multiple separation process scenarios.